MATH 152 **EXAM IIA** NAME Solutions

Mrs. Bonny Tighe 50 points

7.7-11.7

PART II TA_____ Wed 10/30/2019

1. (10 points) Complete the following tests using no abbreviations.

GEOMETRIC SERIES $\sum_{n=0}^{\infty} a_n (r)^n$ is Convergent if $|r| < 1$

THE COMPARISON TEST Suppose $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms.

i) If $\sum_{n=1}^{\infty} b_n$ is convergent and $a_n \leq b_n$ for all n, then $\sum_{n=1}^{\infty} a_n$ is also convergent

ii) If $\sum_{n=1}^{\infty} b_n$ is divergent and $a_n \geq b_n$ for all n, then $\sum_{n=1}^{\infty} a_n$ is also divergent

THE LIMIT COMPARISON TEST

Suppose that $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms . If $\lim_{n\to\infty} \frac{a_n}{b_n} = c$ where c is >0 and finite, then either both series converge or both diverge.

Name the test: Test for Divergence **If** $\lim_{n\to\infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ is divergent

THE INTEGRAL TEST Suppose f is a positive, decreasing and continuous function on $[1, \infty)$ and let $a_n = f(n)$ then the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if $\int_a^{\infty} f(n)\,dn$ is convergent.

P-SERIES $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is divergent if $p \leq 1$

RATIO TEST

If $\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = L$

$L < 1$ then $\sum a_n$ is absolutely convergent

$L > 1$ then $\sum a_n$ is divergent

$L = 1$ then the test is inconclusive

ROOT TEST

If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L$ and

$L < 1$ then $\sum a_n$ is absolutely convergent

$L > 1$ then $\sum a_n$ is divergent

$L = 1$ then the root test is inconclusive

THE ALTERNATING SERIES TEST

If i) if $b_{n+1} \leq b_n$ for all n and ii) $\lim_{n\to\infty} b_n = 0$ Then $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ is Convergent

REMAINDER ESTIMATE FOR THE INTEGRAL TEST

Suppose $a_n = f(n)$ where f is a continuous, positive, decreasing function for $x \geq n$, and $\sum a_n$ is convergent . If $R_n = S - S_n$, then $\int_{n+1}^{\infty} f(x)\,dx \leq R_n \leq \int_n^{\infty} f(x)\,dx$

2. (10 points) Use (a) the Midpoint Rule (b) the Trapezoidal Rule and (c) Simpson's Rule to approximate the given integral with the specified value of n. Find the error bounds for Simpson's rule. Suppose $|f''''(x)| \le K$ *for* $a \le x \le b$, then

$$|E_S| \le \frac{K(b-a)^5}{180n^4} \qquad \int_1^3 x\sqrt[3]{x}\,dx, \quad n=4 \qquad \int_1^3 x^{4/3}\,dx$$

$5/4 \quad 7/4 \quad 9/4 \quad 11/4$

$1 \quad 3/2 \quad 2 \quad 5/2 \quad 3$

$\Delta x = 1/2$

$$M_4 = \frac{1}{2}\left[(5/4)^{4/3} + (7/4)^{4/3} + (9/4)^{4/3} + (11/4)^{4/3}\right]$$

$$T_4 = \frac{1}{4}\left[1 + 2(3/2)^{4/3} + 2(2)^{4/3} + 2(5/2)^{4/3} + (3)^{4/3}\right]$$

$$S_4 = \frac{1}{6}\left[1 + 4(3/2)^{4/3} + 2(2)^{4/3} + 4(5/2)^{4/3} + (3)^{4/3}\right]$$

$$K = \frac{40}{81}$$

$$f(x) = x^{4/3}$$
$$f'(x) = \frac{4}{3}x^{1/3}$$
$$f''(x) = \frac{4}{9}x^{-2/3}$$
$$f'''(x) = \frac{-8}{27}x^{-5/3}$$
$$f''''(x) = \frac{40}{81}x^{-8/3}$$

largest when $x=1$

$$\boxed{|E_S| \le \frac{40(2)^5}{81(180)4^4}}$$

How large should we take n in order to guarantee that Simpson's approximation for $\int_1^3 x\sqrt[3]{x}\,dx, \quad n=4$ is accurate to within 0.001?

$$\frac{40}{81}\frac{(32)}{180\,n^4} \le \frac{1}{10^3}$$

$$\frac{10^3(40)(32)}{81(180)} \le n^4$$

$$n \ge \sqrt[4]{\frac{1000(40)(32)}{81(180)}}$$

rounded up to the next even integer

3. (10 points) Test the following series for absolute convergence, conditional convergence or divergence. State the test you are using and show all your work.

a) $\sum_{n=1}^{\infty} \frac{n+1}{n^2+3^n}$

Compare to $\sum_{n=1}^{\infty} \frac{1}{3^n}$ a convergent geometric series $|r| = 1/3 < 1$

Limit Comparison Test

$\lim_{n\to\infty} \left(\frac{1}{3^n} \cdot \frac{3^n+n^2}{n+1} \right) \to \frac{\infty}{\infty}$ L'H

$= 1 > 0$ and finite so

$\sum_{n=1}^{\infty} \frac{n+1}{n^2+3^n}$ is also absolutely convergent

b) $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+3}$ $b_n = \frac{1}{2n+3}$

Compare to $\sum_{n=1}^{\infty} \frac{1}{n}$ a divergent p-series $p = 1 \leq 1$

Limit Comparison Test

$\lim_{n\to\infty} \left(\frac{1}{n} \cdot \frac{2n+3}{1} \right) = 2 > 0$ and finite so

$\sum_{n=0}^{\infty} \frac{1}{2n+2}$ is also divergent

i) $b_{n+1} \leq b_n$

ii) $\lim_{n\to\infty} \left(\frac{1}{2n+3} \right) = 0$

so $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+3}$ is conditionally convergent

c) $\sum_{n=1}^{\infty} \frac{4 \cdot 7 \cdot 10 \cdots (3n+1)}{4^n n!}$

Ratio Test

$\lim_{n\to\infty} \left| \frac{4 \cdot 7 \cdot 10 \cdots (3n+1)(3n+4)}{4 \cdot 4^n (n+1) n!} \cdot \frac{4^n n!}{4 \cdot 7 \cdot 10 \cdots (3n+1)} \right|$

$\lim_{n\to\infty} \left| \frac{3n+4}{4(n+1)} \right| = 3/4 < 1$ so

absolutely convergent

4. (10 points) Determine whether the series is convergent or divergent and if it is convergent **find the sum**.

a) $\sum_{n=1}^{\infty} \frac{1}{n^2+2n}$

$\frac{A}{n} + \frac{B}{n+2} = \frac{1}{n(n+2)}$

$A(n+2) + Bn = 1$

$n=0 \quad A = \frac{1}{2}$

$n=-2 \quad B = -\frac{1}{2}$

$\sum_{n=1}^{\infty} \left(\frac{1/2}{n} - \frac{1/2}{n+2} \right)$

$\frac{1/2}{1} + \frac{1/2}{2} + \frac{1/2}{3} + \frac{1/2}{4} + \frac{1/2}{5} + \frac{1/2}{6}$

$- \frac{1/2}{3} - \frac{1/2}{4} - \frac{1/2}{5} - \frac{1/2}{6}$

$\frac{1}{2} + \frac{1}{4} = 3/4$ is the sum $\therefore \sum_{n=1}^{\infty} \frac{1}{n^2+2n}$ is convergent because I could find the sum.

b) $\sum_{n=1}^{\infty} \frac{(-\pi)^n}{e^{2n}}$

$\sum_{n=1}^{\infty} \frac{-\pi(-\pi)^{n-1}}{e^2(e^2)^{n-1}}$

$|r| = \left|\frac{-\pi}{e^2}\right| < 1$ convergent geometric series so the

Sum $= \frac{-\pi/e^2}{1-(-\pi/e^2)}$

$\frac{-\pi}{e^2+\pi}$

5. (5 points). Use the **Integral test** to determine if the series is divergent or convergent.

$\sum_{n=0}^{\infty} \frac{e^n}{1+e^{2n}}$

$f(n) = \frac{e^n}{1+e^{2n}}$ is continuous, decreasing and positive for $n \geq 0$

$\lim_{t\to\infty} \int_0^t \frac{e^n}{1+e^{2n}}\,dn \qquad u = e^n, \; du = e^n dn \qquad \lim_{t\to\infty} \tan^{-1}(e^n)\Big]_0^t$

$\lim_{t\to\infty} \left[\tan^{-1}(e^t) - \tan^{-1}(e^0) \right] = \pi/2 - \pi/4 = \pi/4$ which is finite

So $\sum_{n=0}^{\infty} \frac{e^n}{1+e^{2n}}$ is absolutely convergent

$r = x$

6. (10 points) The given curve is rotated about the y-axis. Find the area of the resulting surface. $S = 2\pi \int_a^b radiusarclength$ $\quad y = \ln(x^2 - 1), \quad 2 \le x \le 3$

dx

$$dy/dx = \frac{2x}{x^2-1}$$

$$(dy/dx)^2 = \frac{4x^2}{(x^2-1)^2}$$

$$SA = 2\pi \int_2^3 x\sqrt{1 + \frac{4x^2}{(x^2-1)^2}}\, dx$$

$$SA = 2\pi \int_2^3 x\sqrt{\frac{(x^2-1)^2+4x^2}{(x^2-1)^2}}\, dx = 2\pi \int_2^3 x\frac{\sqrt{(x^4-2x^2+1+4x^2)}}{x^2-1}\, dx$$

$$2\pi \int_2^3 \frac{x\sqrt{x^4+2x^2+1}}{x^2-1}\, dx = 2\pi \int_2^3 \frac{x\sqrt{(x^2+1)^2}}{x^2-1}\, dx$$

$$2\pi \int_2^3 \frac{x(x^2+1)}{x^2-1}\, dx = 2\pi \int_2^3 \frac{x^3+x}{x^2-1}\, dx$$

$$\begin{array}{r} x \\ x^2-1 \overline{\big) x^3+x} \\ -(x^3-x) \\ \hline 2x \end{array}$$

$$2\pi \int_2^3 x + \frac{2x}{x^2-1}\, dx$$

$$2\pi \left[\frac{1}{2}x^2 + \ln|x^2-1|\right]_2^3 =$$

$$2\pi \left[\left(\frac{9}{2} + \ln|8|\right) - (2 + \ln|3|)\right]$$

$$2\pi \left[5/2 + \ln|8/3|\right]$$

MATH 152
Mrs. Bonny Tighe

EXAM II

50 points
7.7-11.7
PART II

NAME Solutions

TA_____ Wed 10/30/2019

1. (10 points) Complete the following tests using no abbreviations.

GEOMETRIC SERIES $\sum_{n=0}^{\infty} a_n (r)^n$ is divergent if $|r| \geq 1$

THE COMPARISON TEST Suppose $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms.

i) If $\sum_{n=1}^{\infty} b_n$ is convergent and $a_n \leq b_n$ for all n, then $\sum_{n=1}^{\infty} a_n$ is also convergent

ii) If $\sum_{n=1}^{\infty} b_n$ is divergent and $a_n \geq b_n$ for all n, then $\sum_{n=1}^{\infty} a_n$ is also divergent

THE LIMIT COMPARISON TEST

Suppose that $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms . If $\lim_{n\to\infty} \frac{a_n}{b_n} = c$ where c is >0 + finite, then either both series converge or both diverge.

Name the test: Test for Divergence If $\lim_{n\to\infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ is divergent

THE INTEGRAL TEST Suppose f is a positive, continuous and decreasing function on $[1, \infty)$ and let $a_n = f(n)$ then the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if $\int_a^{\infty} f(n)\,dn$ is convergent.

P-SERIES $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent if $p > 1$

RATIO TEST

If $\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = L$

$L < 1$ then $\sum a_n$ is absolutely convergent
$L > 1$ then $\sum a_n$ is divergent
$L = 1$ then the test is inconclusive

ROOT TEST

If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L$ and

$L < 1$ then $\sum a_n$ is absolutely convergent
$L > 1$ then $\sum a_n$ is divergent
$L = 1$ then the root test is inconclusive

THE ALTERNATING SERIES TEST

If i) if $b_{n+1} \leq b_n$ for all n and ii) $\lim_{n\to\infty} b_n \neq 0$ Then $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ is Divergent

REMAINDER ESTIMATE FOR THE INTEGRAL TEST

Suppose $a_n = f(n)$ where f is a continuous, positive, decreasing function for $x \geq n$, and $\sum a_n$ is convergent . If $R_n = S - S_n$, then $\int_{n+1}^{\infty} f(x)\,dx \leq R_n \leq \int_n^{\infty} f(x)\,dx$

2. (5 points). Use the **Integral test** to determine if the series is divergent or convergent.

$$\sum_{n=0}^{\infty} \frac{e^n}{e^{2n}+1}$$

$f(n) = \frac{e^n}{e^{2n}+1}$ which is continuous, positive and decreasing for $n \geq 0$.

$$\lim_{t\to\infty} \int_0^t \frac{e^n}{e^{2n}+1}\,dn = \lim_{t\to\infty} \left[\tan^{-1} e^n\right]_0^t =$$

$u = e^n$
$du = e^n\,dn$

$$\lim_{t\to\infty} \left[\tan^{-1} e^t - \tan^{-1} e^0\right] =$$

$\pi/2 - \pi/4 = \pi/4$ finite limit

So Both $\int_0^\infty \frac{e^n}{e^{2n}+1}\,dn$ and $\sum_{n=0}^{\infty} \frac{e^n}{e^{2n}+1}$ are absolutely converges

3. (10 points) Determine whether the series is convergent or divergent and if it is convergent **find the sum**.

a) $\sum_{n=1}^{\infty} \frac{1}{n^2+3n+2}$

$$\frac{A}{n+1} + \frac{B}{n+2} = \frac{1}{(n+1)(n+2)}$$

$A(n+2) + B(n+1) = 1$

$n=-1 \quad A = 1$

$n=-2 \quad B = -1$

$$\sum_{n=1}^{\infty} \left[\frac{1}{n+1} - \frac{1}{n+2}\right]$$

$\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5}$

$-\frac{1}{3} - \frac{1}{4} - \frac{1}{5}$

$\boxed{\text{Sum} = 1/2}$ Because I could find the sum

$\sum_{n=1}^{\infty} \frac{1}{n^2+3n+2}$ is absolutely convergent

b) $\sum_{n=1}^{\infty} \frac{\pi^{n+1}}{2^{2n}}$ $\quad \sum_{n=1}^{\infty} \frac{\pi \cdot \pi^n}{4^n}$

$$\sum_{n=1}^{\infty} \frac{\pi^2}{4}\left(\frac{\pi}{4}\right)^{n-1}$$

$|r| = \left|\frac{\pi}{4}\right| < 1$ so convergent geometric series + we can find the sum

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}$$

$$\frac{\frac{\pi^2}{4}}{1-\pi/4} = \frac{\pi^2}{4-\pi}$$

4 (10 points) Test the following series for absolute convergence, conditional convergence or divergence. State the test you are using and show all your work.

3pts a) $\sum_{n=1}^{\infty} \frac{1}{n+5^n}$

Compare to $\sum_{n=1}^{\infty} \frac{1}{5^n}$ a convergent geometric series $|r| = \frac{1}{5} < 1$.

Limit Comparison Test

$\lim_{n\to\infty} \left(\frac{1}{5^n} \cdot \frac{5^n+n}{1} \right) = L'H$

$\lim_{n\to\infty} \left(\frac{5^n \ln 5 + 1}{5^n \ln 5} \right)$ $L'H$

$\lim_{n\to\infty} \frac{5^n (\ln 5)^2}{5^n (\ln 5)^2} = 1 > 0$ and finite so

$\sum_{n=1}^{\infty} \frac{1}{n+5^n}$ is also absolutely convergent

4pts b) $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n^2+2n+1}}$ $b_n = \frac{1}{\sqrt{n^2+2n+1}}$

Compare to $\sum_{n=1}^{\infty} \frac{1}{n}$ a divergent p-series $p = 1 \le 1$

Limit Comparison Test

$\lim_{n\to\infty} \left(\frac{1}{n} \cdot \frac{\sqrt{n^2+2n+1}}{1} \right) = 1 > 0$ & finite

so $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n^2+2n+1}}$ is also divergent.

Now use Alternating Series Test

$b_n = \frac{1}{\sqrt{n^2+2n+1}}$

i) $b_{n+1} \le b_n$ ✓

ii) $\lim_{n\to\infty} \frac{1}{\sqrt{n^2+2n+1}} = 0$ so

$\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n^2+2n+1}}$ is Conditional Convergent

3pts c) $\sum_{n=1}^{\infty} \frac{(3\cdot5\cdot7\cdot9\cdots(2n+1))}{3^n\, n!}$

Ratio Test

$\lim_{n\to\infty} \left| \frac{(3\cdot5\cdot7\cdot9\cdots(2n+1)(2n+3))}{3\cdot3^n (n+1)\, n!} \cdot \frac{3^n n!}{(3\cdot5\cdot7\cdot9\cdots(2n+1))} \right|$

$\lim_{n\to\infty} \left| \frac{2n+3}{3(n+1)} \right| = 2/3 < 1$ so Absolutely Convergent

$r = y$

5. (10 points) The given curve is rotated about the x-axis. Find the area of the resulting surface. $S = 2\pi \int radius\ (arclength)$ $\quad x = \ln(y^2 - 1), \quad 2 \le y \le 3$

dy

$$S = 2\pi \int_2^3 y \sqrt{1 + (dx/dy)^2}\, dy$$

$$dx/dy = \frac{1}{y^2-1}(2y)$$

$$(dx/dy)^2 = \frac{4y^2}{(y^2-1)^2}$$

$$S = 2\pi \int_2^3 y \sqrt{1 + \frac{4y^2}{(y^2-1)^2}}\, dy$$

$$S = 2\pi \int_2^3 y \sqrt{\frac{(y^2-1)^2 + 4y^2}{(y^2-1)^2}}\, dy$$

$$S = 2\pi \int_2^3 y \sqrt{\frac{y^4 - 2y^2 + 1 + 4y^2}{(y^2-1)^2}}\, dy = 2\pi \int_2^3 y \frac{\sqrt{y^4 + 2y^2 + 1}}{y^2-1}\, dy$$

$$2\pi \int_2^3 y \frac{\sqrt{(y^2+1)^2}}{y^2-1}\, dy = \boxed{2\pi \int_2^3 y \frac{(y^2+1)}{y^2-1}\, dy}$$

5pts

$$\begin{array}{r} y \\ y^2-1 \overline{\big) y^3 + y} \\ -(y^3 - y) \\ \hline 2y \end{array}$$

$$2\pi \int_2^3 y + \frac{2y}{y^2-1}\, dy$$

$$S = 2\pi \left[\tfrac{1}{2}y^2 + \ln|y^2-1| \right]_2^3$$

$$= 2\pi \left[\left(\tfrac{9}{2} + \ln|8|\right) - \left(2 + \ln|3|\right) \right]$$

$$= 2\pi \left[\tfrac{5}{2} + \ln|8/3| \right]$$

6. (10 points) Use (a) the Midpoint Rule (b) the Trapezoidal Rule and (c) Simpson's Rule to approximate the given integral with the specified value of n. Find the error bounds for Simpson's rule. Suppose $|f''''(x)| \le K$ *for* $a \le x \le b$, then

7pts

$$|E_S| \le \frac{K(b-a)^5}{180n^4} \qquad \int_1^2 x\sqrt[3]{x}\,dx, \quad n = 4$$

$\Delta x = 1/4$

$9/8 \quad 11/8 \quad 13/8 \quad 15/8$

$1 \quad 5/4 \quad 3/2 \quad 7/4 \quad 2$

$$M_4 = \frac{1}{4}\left[\frac{9}{8}\sqrt[3]{9/8} + 11/8\sqrt[3]{\tfrac{11}{8}} + 13/8\sqrt[3]{13/8} + 15/8\sqrt[3]{15/8}\right]$$

$$T_4 = \frac{1}{8}\left[1 + 2\left(\tfrac{5}{4}\sqrt[3]{5/4}\right) + 2\left(\tfrac{3}{2}\right)\sqrt[3]{3/2} + 2\left(\tfrac{7}{4}\right)\sqrt[3]{7/4} + 2\sqrt[3]{2}\right]$$

$$S_4 = \frac{1}{12}\left[1 + 4(5/4)\sqrt[3]{5/4} + 2(3/2)\sqrt[3]{3/2} + 4(7/4)\sqrt[3]{7/4} + 2\sqrt[3]{2}\right]$$

$f(x) = x^{4/3}$

$f'(x) = 4/3\, x^{1/3}$

$f''(x) = \frac{4}{9} x^{-2/3}$

$f'''(x) = \frac{-8}{27} x^{-5/3}$

$f''''(x) = \left|\frac{40}{81} x^{-8/3}\right|$ largest at $x = 1$ $\quad K = \frac{40}{81}$

$$|E_S| \le \frac{40}{81}\,\frac{15}{180(4)^4} \le \frac{40}{81(180)4^4}$$

How large should we take n in order to guarantee that the Simpson's rule approximation for $\int_1^2 x\sqrt[3]{x}\,dx$, $n = 4$ is accurate to within 0.01?

3pts

$$\frac{40}{81(180)n^4} \le \frac{1}{100}$$

$$\frac{4000}{81(180)} \le n^4$$

$$\sqrt[4]{\frac{4000}{81(180)}} \le n$$ the next larger even integer